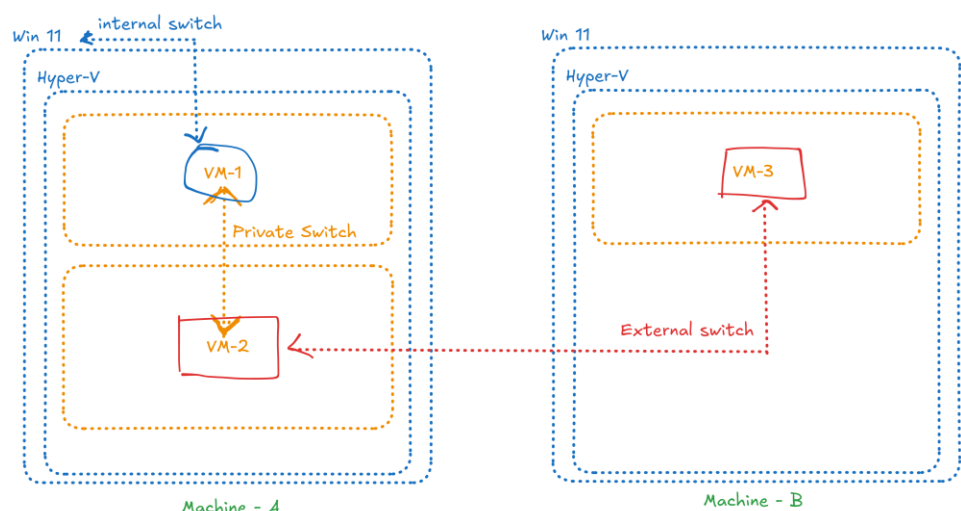


Microsoft Hyper-V:

- Hyper-V is a Type-1 and Type-2 Hypervisor.
 - o Server → Windows Server Hyper-V Edition
 - o Client → Windows 10/11 (Pro or Enterprise edition)
 - We can enable Hyper-V on client by using control panel (Turn Windows feature on or off).
- Components of Hyper-V
 - o *Hypervisor*
 - The main virtualization layer that runs the VMs.
 - o *VMMS*
 - Virtual Machine Management Service
 - This service manages the VMs lifecycle like start the VM, stop the VM and so on.
 - o *VHD/VHDX*
 - Formats (extension) of a virtual HDD.
 - VHD
 - o Maximum supported HDD: 2TB
 - o Generation 1 VM.
 - VHDX
 - o Maximum supported HDD: 64TB
 - o Generation 1 VM.
 - o *Virtual Switch*
 - It's a networking component allows the connectivity within VMs and Host machine.
 - Virtual switches are of 3 types in Hyper-V:
 - *Private Switch*
 - o Allow communication between 2 or more VMs only within a single Hypervisor.
 - *Internal switch*
 - o Allow communication between VMs and host machine within a single Hypervisor.
 - *External switch*
 - o Allow communication from the VM to internet or another Hyper-V on a different computer within a LAN.



- o *Checkpoint*
 - Snapshots (in VMWare workstation).
- o *Integration services*
 - For enhancement of the VM performance (similar to VMWare tools).

- ✓ **VMI Provider**
 - Virtual Machine Interface Provider
 - VMI provider is a WMI-based (windows management instrumentation) interface that enables the management and automation of Hyper-V via powershell scripts.
- ✓ **VM worker process**
 - Aka “vmwp.exe”.
 - It is responsible for managing the state, input/output and execution of the VM.
- ✓ **VSP**
 - Virtual Service Provide
 - This handles the communication and resource sharing with throughout the VMBus.
- ✓ **VMBus**
 - It is a high-speed communication channel.
- ✓ **VSC**
 - Virtualization service consumer
 - It's a guest VM component that communicates with the host's storage, memory and network.

Key features:

- Virtual machine:
 - It supports linux and windows VMs.
 - It supports dynamic memory allocation.
 - NUMA (Non-Uniform Memory Access) architecture.
- It supports networking switches
 - Internal
 - External
 - Private
- Live Migration
 - Hot-migration
 - No downtime, means the VM will not be powered-off.
- Storage Migration
 - Moving the storage between physical locations, while VMs are running.
- Dynamic memory
 - Allocates the RAM to the VM as needed.
- Shielded VMs
 - Only for windows servers
 - It encrypts the virtual machine.

System requirements:

- 64bit processor
- Intel-VT (Virtualization Technology) or AMD-V must be “enabled”.
- Minimum 4GB RAM
- Recommended 8GB RAM
- Windows 10/11 pro or enterprise edition must be used for client Hyper-V
- Windows server 2012R2 or above for server Hyper-V

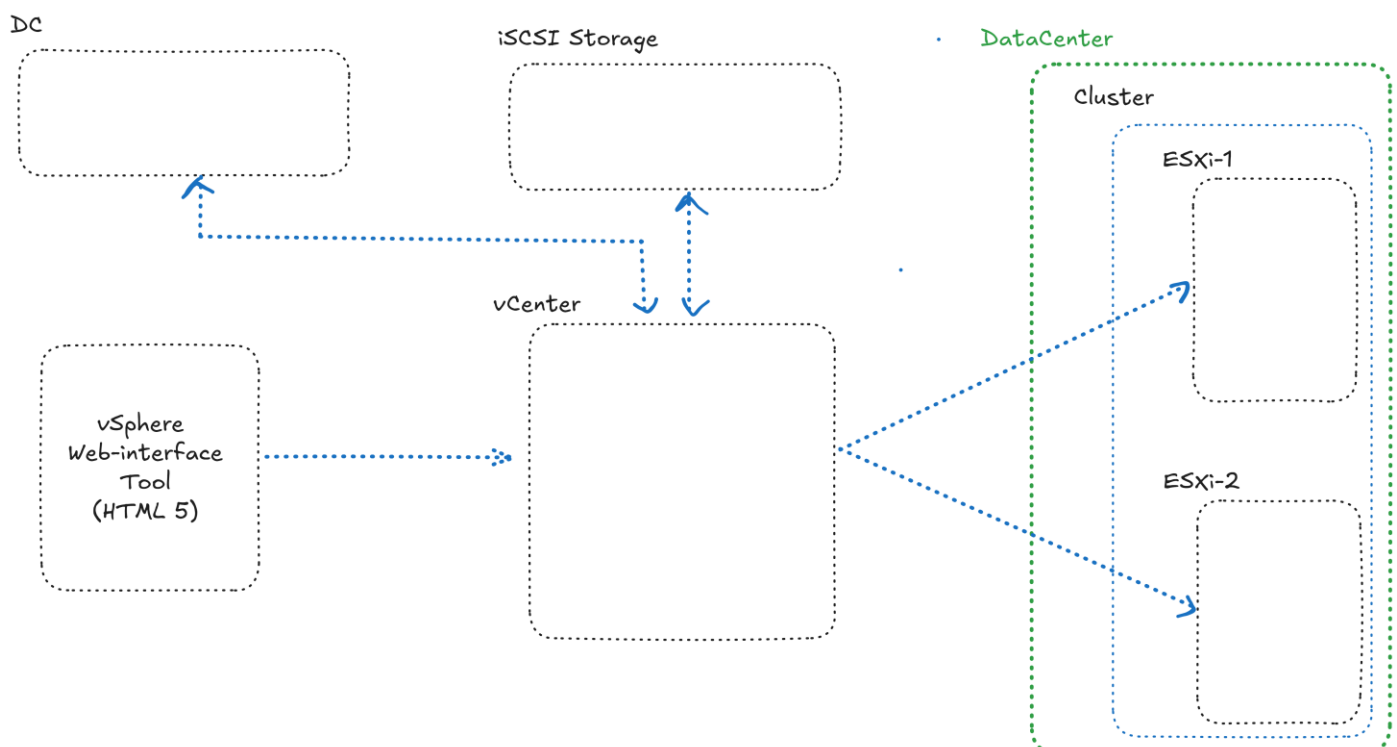
VMWare vSphere Environment:

- VMWare vSphere is an enterprise-grade virtualization platform used to build and manage *virtualized IT infrastructure*.
- It allows organizations to run, manage and secure applications in virtual machines across a pool of hardware.
- VMWare vSphere has 2 major components
 - ESXi server (*Elastic Sky X Integrated*)
 - It is Type-1 hypervisor which is installed directly on physical hardware (bare-metal).
 - It is required to host and run virtual machines.
 - vCenter server
 - This is a centralized management platform for managing multiple ESXi hosts/server and their VMs.
 - vCenter server offers features like:
 - vMotion (migration)
 - DRS (Distributed Resource Scheduler)
 - HA (High Availability)
 - FT (Fault Tolerance)
 - vSAN
 - vSphere client tool (exe)
 - Deprecated after version vCenter server 6.5
 - Now we have web browser-based tool)

Key components/features:

- ✓ **vMotion** – live migration of running VMs between 2 ESXi servers with no downtime.
- ✓ **Storage vMotion** – migration of VMs HDD/files from one datastore (storage) to another without downtime.
- ✓ **High Availability** – automatically restart/connecting VMs on another ESXi hosts, if one fails.
- ✓ **DRS** – Balances workloads across hosts for optimal performance.
- ✓ **vSAN** – enterprise grade SAN storage solution.
- ✓ **Templates** – it is the pre-configured image to create more similar VMs.
- ✓ **Clones** – you can create identical copy of a VM.

VMWare vSphere architecture:



ESXi

- The work of ESXi is to run VMs on top of it.
- Minimum requirements for ESXi:
 - o 2 core processors
 - o Intel-VT/AMD-V must be enabled.
 - o 4GB RAM
 - o 32GB HDD
 - o 1 Gbps ethernet adapter
 - o Bootable device (USB, HDD, SSD)
- When ESXi is installed, a switch gets created (virtual Standard switch (vSS)) which helps in the communication between user/vCenter to ESXi.

vCenter server

- The job of vCenter is to
 - o Create VMs, storage, network, etc...
 - o Manage the VMs
 - o Connect with ESXi server centrally.
 - o Manage ESXi using Virtual Distributed Switch (vDS)
 - o Minimum requirements for vCenter server:
 - 2 vCPUs
 - 12 GB RAM
 - 415GB HDD
 - Maximum ESXi hosts support up to: 10 ESXi hosts (Tiny)
 - Maximum Virtual Machine support up to: 100 VMs (Tiny)
- vCenter can be deployed in 2 ways:
 - o Windows-based vCenter server
 - Deprecated since vSphere 7.0
 - o *vCenter Server Appliance (VCSA)*
 - recommended one.
 - This is Linux-Based (Photon OS, by VMware).

vSphere web-interface tool

- A web-based interface (HTML 5) for interacting (GUI) with ESXi and vCenter server.
- This interface can access vCenter or ESXi from any GUI system (DC, Node01, Node02) within the network.

iSCSI storage

- This is one of the supported storages for vCenter's centralized datastore.
- Other supported storages are:
 - o NFS
 - o vSAN
 - o local datastore
 - o iSCSI storage
- this iSCSI storage will be created on DC machine and will be accessed on both ESXi servers via vCenter.

DC

- vCenter requires domain controller to connect and authenticate.
- We can create users in DC and give them various access over vCenter server.